Classes

CSC 240 Data Structures Fall 2025

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1 Objectives

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4. ☒ Member functions
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6. ☒ Constructor and operator overloading
7. ☒ Separation of interface and implementation

Other topics:

1. ☐ Arrays of Objects
2. ☐ Simulating dice (case study)

3. ☐ Unified Modeling Language (UML)
4. ☐ Instance and static members
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8. ☐ Object Conversion
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13. ☐ Inheritance (derivative classes)
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15. ☐ Virtual Functions (overriding member functions at runtime)

2 Procedural vs. object-oriented programming (OOP) paradigms

- Most disciplines that grow over time are not just cobbled together: People identify patterns that are successful, explore and pursue them, and finally turn them into standards (until they fail).
- These patterns are also called "paradigms" (). Two important programming paradigms are procedural and object-oriented programming.
- What does **procedural** mean? What does it achieve?

  > Centered creating procedures or functions. The main purpose is to create modules and reuse them over and over again.

- What is **object orientation**? What does it achieve?

OO means that everything is an object. And every object has its own state and behavior. Computation now happens by message (data) exchange between objects - like a conversation between agents.

Encapsulation of data (withhold some data), inheritance (of objects), polymorphism (operator overloading at compile-time).

3 Example: A light switch as message exchange between objects

- Example 1: A light switch.

```
#include <iostream>
using namespace std;

// Let there be light!
class Light {
public:
  void turnOn() { cout << "Light is on.\n"; }
  void turnOff() { cout << "Light is off.\n"; }
};
// Provide a switch!
class Switch {
  Light *light; // knows where to send the message
public:
  Switch(Light *l) : light(l) {}; // flips the switch
  void flip(bool on) {
    if (on)
      light->turnOn(); // sends 'turnOn' message to the Light
    else
      light->turnOff(); // sends 'turnOff' message to the Light
  }
};
// Test it by switching light on and off!
int main()
{
  Light lamp; // create a Light object called 'lamp'
  Switch s(&lamp); // create a Switch for the lamp
```

```
  s.flip(true); // throw the switch and turn lamp on
  s.flip(false); // throw the switch and turn lamp off
  return 0;
}
```

- Explanation:

  The `Switch` does not directly manipulate the `Light` - it sends messages `turnOn` and `turnOff`. Each object controls its own behavior and data - this is a huge advantage when you want to solve more and more difficult problems.

- **Challenge (bonus assignment):**

  What would the procedural version of this program look like?

4 Example: Procedural vs. object-oriented adding of objects

- What is **object-oriented** programming?

  OOP is centered on creating and manipulating objects in a safe way:

- Example using an `add` function in procedural style:

  1. You may want to add integers.
  2. You may want to add floating-point numbers.
  3. You may want to "add" (concatenate) strings.

- These are three different procedures. In C/C++, they can be **overloaded** so that each function is called `add` but which one is used depends on its arguments.

- Implementation:

```
#include <iostream>
using namespace std;

// Integer addition
int add(int a, int b)
```

```
{
  return (a+b);
}

// Floating-point addition
double add(double a, double b)
{
  return (a+b);
}

// String concatenation
string add(std::string a, std::string b)
{
  return (a+b);
}
// The main program calls each function
int main()
{
  cout << add(5,7) << endl; // addition of two integers
  cout << add(5.5,6.5) << endl;  // addition of two floats
  cout << add("five","seven") << endl; // concatenation of two strings
  return 0;
}
```

- In the OOP style, a `class` named `Adder` has member functions that take care of both argument types:

```
#include <iostream>
using namespace std;

// The Adder class has 'public' methods (no data)
class Adder {
public:
  int add(int a, int b) // integer addition
  {
    return (a+b);
  }
  double add(double a, double b) // double addition
  {
    return (a+b);
```

```
  }
  string add(std::string a, std::string b) // string 'addition'
  {
    return (a+b);
  }
};
// The functions are called on the Adder object A
int main()
{
  Adder A;
  cout << A.add(5,7) << endl;
  cout << A.add(5.5,6.5) << endl;
  cout << A.add("five","seven") << endl;
  return 0;
}


12
12
fiveseven
```

- Problems with Procedural Style
  - All `add` functions live in global scope.
  - Namespace pollution: more name conflicts.
  - Maintenance gets harder.
  - There are no objects and no states.
- Advantage of OOP
  - All `add` methods are now grouped in one class `Adder`.
  - Code is encapsulated (easier to read, find, reason about).
  - Behavior can be extended - `reset`, `showState`
  - You can associate behavior with state - `count`, `lastResult`.
- Criticisms of OOP
  - Significant overhead if you have class hierarchies.
  - The code can look more tangled
  - When the logic is simple then OOP adds needless structure.

  - Encapsulation can be illusion.
  - Tight couping - classes are very interdependent
  - Testing can be difficult

- Why do "objects" and their "state" matter?

  - Object mimic real identities: `BankAccount` has data like `balance` and methods like `deposit`, `withdraw`, `checkBalance`.
  - Reuse objects and evolve them by extending your classes.
  - In UI (Amazon checkout) and in games, OOP rules.

5 An everyday "data encapsulation" example: The car

- What's part of the car's "user interface"? (What can you see/use?)

  To use a car (as driver = user):

  - turn the wheel
  - open/close doors
  - use the gas pedal
  - brake
  - gear shift
  - display data on a dashboard

- What's part of the car's "internal implementation"? (What's hidden?)

  Mechanisms hidden from the user (to be able to use the car):

  - Steering column
  - Fuel injection
  - Brake system
  - Most things outside of the driver's field of vision

6 Classes and objects

- What is a `class`? A `class` is three things: **keyword**, **code**, and **concept**.

- As a **keyword**:

It's a type that you can use to cast data - you can make your own categories. `Light` class allowed us to define a `lamp`.

- As **code**:

  A `class` encapsulates the attributes (properties, member data), and behavior (member functions, methods) that any object of that class may have. When we define a `lamp` as a `Light` we can now turn it on/off.

```
// The Rectangle class objects have width and length
// You can set and get these properties for any Rectangle
class Rectangle {
private:  // member data = properties
  double width;
  double length;
public: // member functions = behavior
  // setter / mutators
  void setWidth(double);
  void setLength(double);
  // getter / accessors
  double getWidth();
  double getLength();
  double getArea();
};
```

- As a **concept:**

  A `class` describes an object like a blueprint

- The blueprint can be used to build any number of objects, or **instances**.

- Example: What could you say about the `Insect` class and the two objects shown in the figure?

  Answer:

  - `Insect` can describe attributes and behaviors of MANY insects.
  - `housefly` and `mosquito` and instantiated from `Insect`
  - `mosquito` members have a `stinger` and they `sting`
  - `housefly` members have a `sucker` and they `suck`

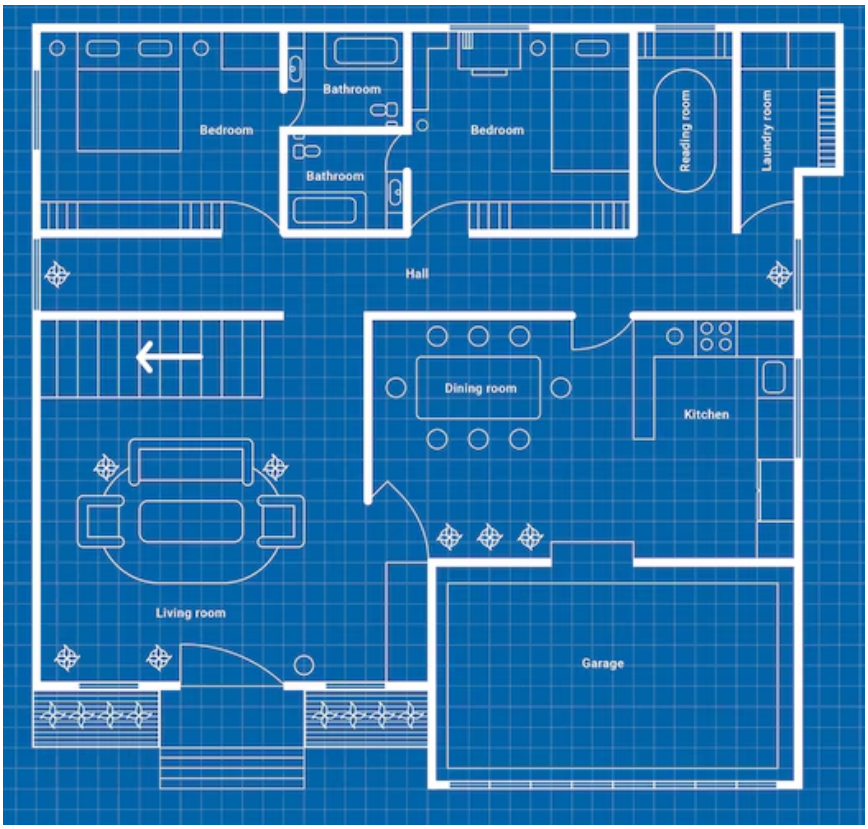


Figure 1: Blueprint of a house.

Figure 2: House objects build with a house blueprint.

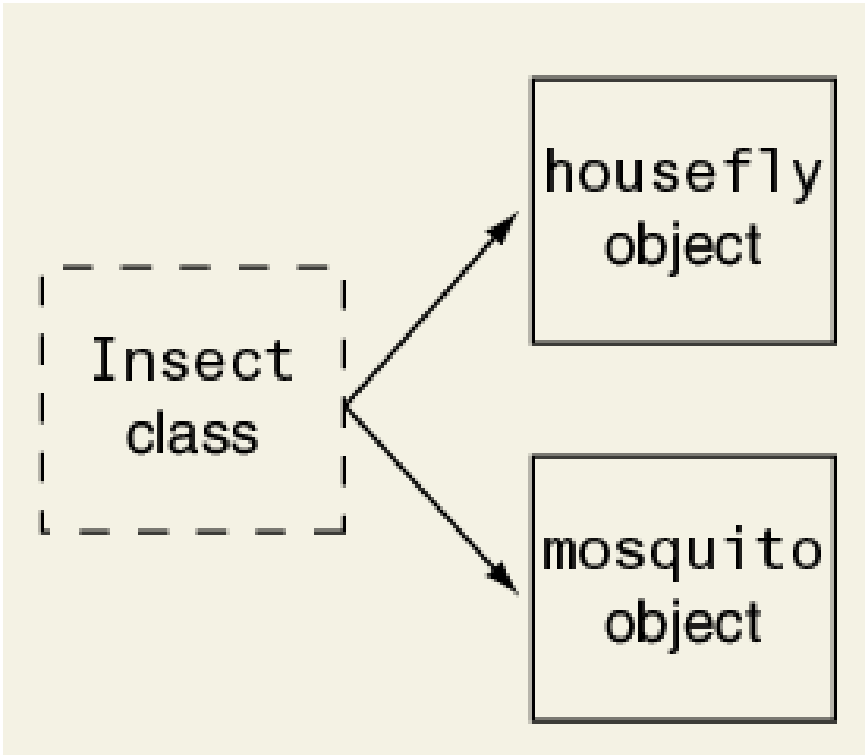


Figure 3: An Insect class and two of its instances.

7 Using the `string` class

- You've already used C++ classes like `vector` and `string`.
- To use the `string` class you need to `#include <string>`. Now you can define `string` objects, and use string member functions.
- Example: Setting a city name, a State, and getting its length.

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
  string cityName;    // create a string class object
  cityName = "Batesville";  // assign string to it
  cout << cityName << endl;
  cout << cityName.length() << endl;
  cout << cityName.append(", AR");
  return 0;
}


Batesville
10
Batesville, AR
```

- What about the time complexity of the `length` and `append` functions?

Operation	Time	Notes
`std::string::length`	O(1)	Length is stored internally at compile time
`std::string::append`	O(1)	Places new characters in allocated space

8 Introduction to classes

Let's use a **bank account** as an example (since everyone has got one).

- What are the computing **procedures** to run a **bank account**?
  - get the balance

  - deposit an amount (set)
  - withdrawing an amount (set/mutate)

- What are the **data** of a bank account?

  - starting balance
  - current balance
  - `private` member data

- What does designing a `BankAccount class` mean?

  1. determine your data (`balance`).
  2. write procedures to manage the data flow.
  3. create `BankAccount` objects in `main`
  4. use the objects exclusively through `public` member functions like `getBalance`, `depositAmount`.

- What's an **object**?

  An object is a resource used to complete the task - no difference to other resources (variables) - region in memory.

  To the `class` the object is an instance of what the class represents - like a `lamp` that is a `Light` object.

- Example of an object: A savings account, and we have a `BankAccount`. What does it have?

  A savings account is a special type of `BankAccount`. Just like 5 is a special integer (`int`).

  1. it has data (`balance`, account number...)
  2. it has behavior/operations (deposit, withdrawal, balance)
  3. it allows you interact with it.
  4. if you want anything special you need to define a subclass

- What's a constructor? Example?

  A constructor constructs an object of the chose class. In `BankAccount`, our constructor creates an account with a starting balance.

9 OOP workflow and class template

- How to do it in practice:

    1. Write a class (category,type)
    2. Create instances of the class (objects)
    3. Use class methods on class data = OOP

- General template:

```
class ClassName {

  private:
     // data members (attributes)

  public:
     // member functions (methods)
};
```

- Member functions can be defined **inline** (inside the `class` template), or externally. Then they have to be brought into **scope** with the scope operator `::`

- Example: Code along using OneCompiler.Com. Insert this code between the `namespace` statement and the `main` program, like a function. Note: All of the member functions are defined **inline**.

```
class BankAccount
{
private:
  double balance;  // data member

public:

  BankAccount(double startingBal)  // constructor
  { balance = startingBal; }

  void deposit(double amount) // setter / mutator function
    { balance += amount; }

  void withdraw(double amount) // setter / mutator function
```

```
    { balance -= amount; }

  double getBalance() const  // constant getter / accessor function
  { return balance; }

};

int main()
{
  BankAccount acct(100); // create BankAccount with 100 starting balance
  cout << "Starting balance: " << acct.getBalance() << endl;
  acct.deposit(100);
  cout << "New balance: " << acct.getBalance() << endl;
  acct.withdraw(50);
  cout << "New balance: " << acct.getBalance() << endl;
  return 0;
}
```

10 Member functions and member data encapsulation

- There are some similarities between defining/using functions (procedures) and classes.
- What if we do not define a member function inside the `class` definition?

  Then we need to define them externally.

- The compiler has to be told that it is part of the `class`:

```
// outside of the class declaration
BankAccount::BankAccount(double startingBal) // with scope resolution
{ balance = startingBal; }
```

- The main differences to the **inline** definition:

  1. The function parameter `startingBal` is no longer optional.
  2. The function needs to be brought into scope with the prefix `BankAccount::`

3. The function definition must be located after the `class` declaration, and before the `main` function.
4. Typically, declarations are put into header files `.h`, while the member functions are in a separate file `.cpp`. The header is given to the preprocessor, and the definitions file is compiled separately (it becomes an `.o` object file) and later linked to the object file for the `main` program.

- What's going on conceptually here?

  This is an example of separating interface and implementation - it mirrors the separation of function prototype and function definition in procedural programming.

11 Using a class in a `main` function

- We must manage the data flow between the `class` definition and the `main` program, where `class` objects are created and used.
- Example with `BankAccount`:
  1. Create a `BankAccount` named `acct` with 100 starting balance.
  2. Show the current balance.
  3. Deposit 10 and show balance.
  4. Withdraw 50 and show balance.
- Code: Use your previous definition of `BankAccount` and change `main`.

```
#include <iostream>
using namespace std;

class BankAccount
{
private:
  double balance;

public:
  BankAccount(double startingBal)
  { balance = startingBal; }
```

```
  void deposit(double amount)
  { balance += amount; }

  void withdraw(double amount)
  { balance -= amount; }

  double getBalance() const
  { return balance; }

};

int main()
{

  return 0;
}
```

12 PRACTICE Separate interface and implementation

- Using the `BankAccount` class provided in the starter code, separate the class interface and the implementation by splitting the code into:
  1. `class` declaration with member data and member function prototypes,
  2. definitions of the member function prototypes.
  3. When you're done, create a `BankAccount` named `account` with a starting balance of 200, deposit 300, withdraw 100, and print the balance each time.
- Sample output:

```
Account starting balance: $200.
Deposited $300. New balance: $500.
Withdrew $100. New balance: $400.
```

- Starter code: tinyurl.com/bank-account-test

```
// The BankAccount class has a balance.  It is created with a starting
```

```
// balance. You can deposit or withdraw funds. You can get a balance
// printout.
class BankAccount
{
private:
  double balance;

public:
  BankAccount(double startingBal)
  { balance = startingBal; }

  void deposit(double amount)
  { balance += amount; }

  void withdraw(double amount)
  { balance -= amount; }

  double getBalance() const
  { return balance; }

};
```

- Solution video: `https://youtu.be/CyJvZ7nB_IU`)

13 Example: A `Rectangle` class

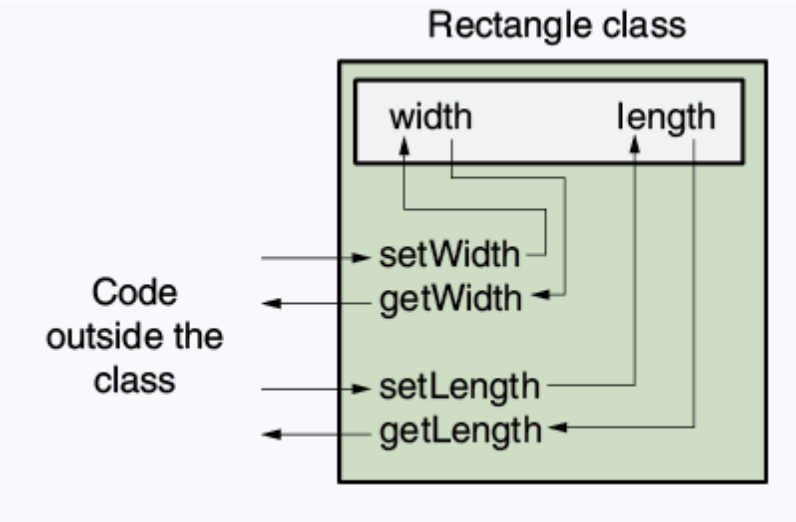
- Which data members should a `Rectangle` class have? Which types?

  As a geometric object it is fully defined by its `width` and its `length`. These should be `double`.

- Which member functions should a `Rectangle` class have?
  - Getters: `getWidth`, `getLength`
  - Setters: `setWidth`, `setLength`
- What `return` types and arguments should the member functions have?
  - Getters: `getWidth`, `getLength` <- `double` (give back member data)

  - Setters: `setWidth`, `setLength` <- `void` (change member data)

- In OOP, an object protects its important data by making them `private`, and provides a `public` interface to access the data.



14 PRACTICE Create a `Rectangle` class

- Using the starter code below, **write a declaration** for the `Rectangle` class that includes the `width` and `length` data members, and the member functions `getWidth`, `getLength`, `setWidth`, and `setLength`.

  Write all member functions as **inline functions** (definitions in the `class` declaration). Remember to **run your code** as soon as you finished a statement for syntax checks.

  Test this class using the `main` function:

  1. Create a `Rectangle` object `r` with width 4 and length 5.
  2. Print `width` and `length` of the object.

- Sample output:

```
Width = 4
Length = 5
```

- Starter code: tinyurl.com/rectangle-class-starter

```
// Declare and test a Rectangle class
// Header files

// Namespace declaration

// Class declaration

// private member data
// rectangle width
// rectangle length
// public member functions

// set rectangle width
// set rectangle length
// get rectangle width
// get rectangle length

// Main function

// create a Rectangle object r
// set object width to 4
// set object length to 5
// show object width
// show object length
```

```
// Declare and test a Rectangle class
// Header files
#include <iostream>
// Namespace declaration
using namespace std;
// Class declaration
class Rectangle {
  // private member data
private:
  double width; // rectangle width
  double length;// rectangle length
  // public member functions
public:
```

```
  // set rectangle width
  void setWidth(double);
  // set rectangle length
  void setLength(double l) { length=l; }
  // get rectangle width
  double getWidth() const { return width; }
  // get rectangle length
  double getLength() const { return length; }
};
// Function definitions
void Rectangle::setWidth(double w)
{
  width=w;
}

// Main function
int main()
{
  // create a Rectangle object r
  Rectangle r;
  // set object width to 4
  r.setWidth(4.);
  // set object length to 5
  r.setLength(5.);
  // show object width
  cout << "Width = " << r.getWidth() << endl;
  // show object length
  cout << "Length = " << r.getLength() << endl;
  return 0;
}
```

15 PRACTICE Separate interface and implementation files

- We separated `class` declaration (interface part) and member function definitions (implementation part) in an earlier exercise.
- Now, we're going to store the declaration in a header file `Rectangle.h`, the definitions in a C++ file `Rectangle.cpp`, and the testing program in a third file, `RectangleTest.cpp`.

- How can we run the test using these three files? Take a guess!

    1. The `Rectangle.h` file (which also contains the `#include` and `namespace` statements) will be included as a header file in both `Rectangle.cpp` and `RectangleTest.cpp`.
    2. `Rectangle.cpp` will be compiled into an object file `Rectangle.o` without linking using the `-c` option of the `g++` (GCC) compiler.
    3. `RectangleTest.cpp`, which contains the `main` function, will be compiled and linked with `Rectangle.o`.
    4. It turns out that the Linux `make` command can execute all of these steps using a configuration file called a `Makefile`.

- We're first going to do this manually on Google Cloud Shell, and then with a `Makefile`.

16 PRACTICE Tangle separate files

- Create `Rectangle.h`

```
#ifndef RECTANGLE_H
#define RECTANGLE_H
#include <iostream>
// Namespace declaration
using namespace std;

// Class declaration
class Rectangle {
  // private member data
private:
  double width; // rectangle widtho
  double length;  // rectangle length
  // public member functions
public:
  void setWidth (double );  // set rectangle width
  void setLength (double ); // set rectangle length
  double getWidth() const;  // get rectangle width
  double getLength() const; // get rectangle length
  double getArea() const;   // compute and get rectangle area
```

```
};
#endif
```

- Create `Rectangle.cpp`

```
#include "Rectangle.h"

// Member function definitions
//***************************************************************
// setWidth assigns its argument to the private member width     *
//***************************************************************
void Rectangle::setWidth (double w)
{
  width = w;
}
//***************************************************************
// setLength assigns its argument to the private member length   *
//***************************************************************
void Rectangle::setLength (double l)
{
  length = l;
}
//***************************************************************
// getWidth returns the value in the private member width        *
//***************************************************************
double Rectangle::getWidth() const
{
  return width;
}
//***************************************************************
// getLength returns the value in the private member length      *
//***************************************************************
double Rectangle::getLength() const
{
  return length;
}
//***************************************************************
// getArea computes and returns the area from private data       *
//***************************************************************
double Rectangle::getArea() const
```

```
{
  return (width * length);
}
```

- Create `RectangleTest.cpp`

```
#include "Rectangle.h"

// main function
int main()
{
  Rectangle box1;
  box1.setWidth(12.7);
  box1.setLength(14.5);
  cout << "Box 1: "
       << box1.getWidth() << " x " << box1.getLength()
       << endl;
  Rectangle box2;
  box2.setWidth(8.5);
  box2.setLength(29.3);
  cout << "Box 2: "
       << box2.getWidth() << " x " << box2.getLength()
       << endl;
  // Test getArea()
  if (box1.getArea() < box2.getArea())
    cout << "Box 1 is smaller.\n";
  else
    cout << "Box 2 is smaller.\n";
  return 0;
}
```

17 PRACTICE Compiling and testing a `class` from multiple files

1. Log into your Lyon Google account and open ide.cloud.google.com
2. Wait until your Linux Virtual Machine has been started (1 min).
3. Ignore the push to use Gemini, the AI assistant, instead just enlarge the terminal window (lower half of the screen).

4. The following commands are useful - enter them after the `$` prompt:

```
$ ll              # alias for ‘ls -alF‘ show details of all files
$ cd dir          # change to directory ‘dir‘
$ cd ..           # go one level up
$ nano file       # open ‘file‘ in the ‘nano‘ editor
$ man cmd         # get documentation on ‘cmd‘
$ g++ file -o out # compile ‘file‘ into object file ‘out‘
$ file file       # give information about ‘file‘
$ clear           # clear the screen
$ cat file        # display the content of ‘file‘
$ rm file         # delete ‘file‘
```

5. Get the three files from GitHub using `wget`. Enter the following commands at the `$` prompt, then check they're there with `ll`.

```
wget -O Rectangle.h tinyurl.com/rectangle-h
wget -O Rectangle.cpp tinyurl.com/rectangle-cpp
wget -O RectangleTest.cpp tinyurl.com/rectangletest-cpp
```

6. Execute the following commands to build the executable `rect`, and then run it in the last step:

```
g++ -c Rectangle.cpp
g++ RectangleTest.cpp Rectangle.o -o rect
./rect
```

   If any of your .cpp or .h files cannot be found, add the current directory (`.`) to your `PATH` with: `export PATH="$PATH:."`

7. Use `wget` to get the `Makefile` from tinyurl.com/rectangle-makefile and store it as `Makefile`.

```
wget -O Makefile tinyurl.com/rectangle-makefile
```

8. Let's take a look at it to understand what it does:

```
rectangle: RectangleTest.o Rectangle.o
g++ RectangleTest.o Rectangle.o -o rectangle

Rectangle.o: Rectangle.cpp Rectangle.h
```

```
g++ -c Rectangle.cpp

RectangleTest.o: RectangleTest.cpp Rectangle.h
g++ -c RectangleTest.cpp

clean:
rm -f *.o rectangle
```

9. To use the `Makefile` with `make`, simply run `make` on the Google shell, and then run the executable `./rectangle` to check the outcome. If you run `make` again now, it will not do anything (no file changed):

```
g++ -c RectangleTest.cpp
g++ -c Rectangle.cpp
g++ RectangleTest.o Rectangle.o -o rectangle
```

10. To see `make`'s magic at work, enter the command `touch Rectangle.o` This will change its timestamp and `make` will get to work. Finally, you can run `make clean` to clean up and remove temporary files.

```
g++ RectangleTest.o Rectangle.o -o rectangle
```

18 Example: Planning a house (multiple `Rectangle` objects)

- Constraints: Includes **??** (needs to be tangled and be available as `../src/Rectangle.cpp`, and `Rectangle.h` must also be there). Later, we will see a verion with dynamical allocation.

- Sample input/output:

```
: What is the kitchen's length? 10
: What is the kitchen's width? 14
: What is the bedroom's length? 15
: What is the bedroom's width? 12
: What is the den's length? 20
: What is the den's width? 30
: The total area of the three rooms is 920
```

- Code:

```
#include <iostream>
#include "Rectangle.cpp"
using namespace std;

//*************************************************************
// Function main
//*************************************************************
int main()
{
  double number;     // number input
  double totalArea;  // total area
  Rectangle kitchen; // kitchen dimensions
  Rectangle bedroom; // bedroom dimensions
  Rectangle den;     // den dimensions

  // Get the kitchen dimensions
  cout << "What is the kitchen's length? ";
  cin >> number; cout << number << endl;    // get the length
  kitchen.setLength(number);                // store in kitchen object
  cout << "What is the kitchen's width? ";
  cin >> number; cout << number << endl;    // get the width
  kitchen.setWidth(number);                 // store in kitchen object

  // Get the bedroom dimensions
  cout << "What is the bedroom's length? ";
  cin >> number; cout << number << endl;    // get the length
  bedroom.setLength(number);                // store in bedroom object
  cout << "What is the bedroom's width? ";
  cin >> number;  cout << number << endl;   // get the width
  bedroom.setWidth(number);                 // store in bedroom object

  // Get the den dimensions
  cout << "What is the den's length? ";
  cin >> number; cout << number << endl;    // get the length
  den.setLength(number);                    // store in den object
  cout << "What is the den's width? ";
  cin >> number; cout << number << endl;    // get the width
  den.setWidth(number);                     // store in den object

  // Calculate the total area of the three rooms
```

```
  totalArea = kitchen.getArea() +
    bedroom.getArea() +
    den.getArea();

  // Display total area of the three rooms
  cout << "The total area of the three rooms is "
       << totalArea << endl;
  return 0;
}
```

- Testing the tangled program (org-babel-tangle) - cursor right after the command, then C-x C-e with the values

```
cd ../src
g++ rectangle2.cpp -o rectangle2
echo "10 14 15 12 20 30" | ./rectangle2
```

```
What is the kitchen's length? 10
What is the kitchen's width? 14
What is the bedroom's length? 15
What is the bedroom's width? 12
What is the den's length? 20
What is the den's width? 30
The total area of the three rooms is 920
```

19 Pointers to `class` Objects

- A `class` is a user-defined type. Its objects are regions of memory.
- You can define a pointer that points at a `class` object. You can then use the `->` operator to call member functions (instead of dot `.`)
- Class object pointers can be used to dynamically allocate objects. Unless you use a **smart pointer**, you have to `delete` the object when you're done.
- Code example using the `Rectangle` class:

```
#include <iostream>
#include "Rectangle.cpp"
```

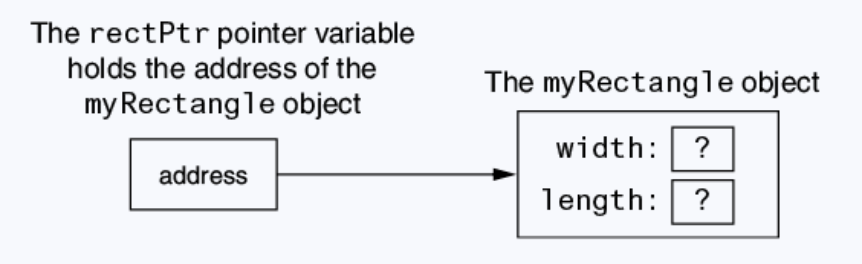


Figure 4: State of the object pointed at before initialization.

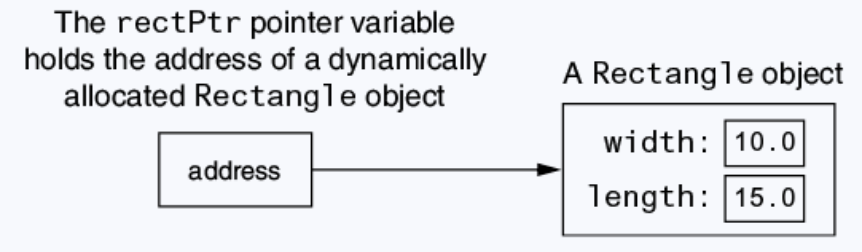


Figure 5: State of the object pointed at after initialization.

```
using namespace std;

int main()
{
  Rectangle R; // create a rectangle
  Rectangle *rectPtr = nullptr; // create a pointer (initialize it)xs
  rectPtr = &R; // pointer points at my rectangle object

  // R.setWidth(12.5); // store width value with the object value
  rectPtr->setWidth(12.5); // store width value with the pointer
  rectPtr->setLength(4.8); // store length value

  cout << "Width: "  << rectPtr->getWidth() << endl;
  cout << "Length: " << rectPtr->getLength() << endl;

  return 0;
}


Width: 12.5
```

```
Length: 4.8


Width: 12.5
Length: 4.8
```

- Same thing with dynamically allocated pointers (`new`, `delete`)

```
#include <iostream>
#include "Rectangle.cpp"
using namespace std;

int main()
{
  Rectangle R; // create a rectangle
  Rectangle *rectPtr = nullptr; // create a pointer (initialize it)xs
  rectPtr = new Rectangle; // dynamically allocate a Rectangle object

  rectPtr->setWidth(10.5); // store width value with the pointer
  rectPtr->setLength(5.8); // store length value

  cout << "Width: "  << rectPtr->getWidth() << endl;
  cout << "Length: " << rectPtr->getLength() << endl;

  // must free allocated memory
  delete rectPtr;
  rectPtr = nullptr;

  return 0;
}


Width: 10.5
Length: 5.8
```

20 PRACTICE Using pointers to objects [Bonus assignment]

- **Task:** Modify the house building program to dynamically allocate `Rectangle` objects, `kitchen`, `bedroom`, and `den`.

- **Constraints:** Includes **??** (needs to be tangled and be available as `../src/Rectangle.cpp`, and `Rectangle.h` must also be there).
- **Sample input/output:**

```
: What is the kitchen's length? 10
: What is the kitchen's width? 14
: What is the bedroom's length? 15
: What is the bedroom's width? 12
: What is the den's length? 20
: What is the den's width? 30
: The total area of the three rooms is 920
```

- **Solution:**

```
#include <iostream>
#include "Rectangle.cpp"
using namespace std;

//***************************************************************
// Function main
//***************************************************************
int main()
{
  double number;                    // number input
  double totalArea;                 // total area
  Rectangle *kitchen = nullptr;     // pointer to Rectangle object
  kitchen = new Rectangle;          // kitchen dimensions
  Rectangle *bedroom = nullptr;     // pointer to Rectangle object
  bedroom = new Rectangle;          // bedroom dimensions
  Rectangle *den = nullptr;         // pointer to Rectangle object
  den = new Rectangle;              // den dimensions

  // Get the kitchen dimensions
  cout << "What is the kitchen's length? ";
  cin >> number; cout << number << endl;     // get the length
  kitchen->setLength(number);                // store in kitchen object
  cout << "What is the kitchen's width? ";
  cin >> number; cout << number << endl;     // get the width
  kitchen->setWidth(number);                 // store in kitchen object
```

```
  // Get the bedroom dimensions
  cout << "What is the bedroom's length? ";
  cin >> number; cout << number << endl;     // get the length
  bedroom->setLength(number);                // store in bedroom object
  cout << "What is the bedroom's width? ";
  cin >> number;  cout << number << endl;    // get the width
  bedroom->setWidth(number);                 // store in bedroom object

  // Get the den dimensions
  cout << "What is the den's length? ";
  cin >> number; cout << number << endl;    // get the length
  den->setLength(number);                   // store in den object
  cout << "What is the den's width? ";
  cin >> number; cout << number << endl;    // get the width
  den->setWidth(number);                    // store in den object

  // Calculate the total area of the three rooms
  totalArea = kitchen->getArea() +
    bedroom->getArea() +
    den->getArea();

  // Display total area of the three rooms
  cout << "The total area of the three rooms is "
       << totalArea << endl;

  // Deallocate memory
  delete kitchen;
  kitchen = nullptr;
  delete bedroom;
  bedroom = nullptr;
  delete den;
  den = nullptr;
  return 0;
}
```

- Testing the tangled program (org-babel-tangle) - cursor right after the command, then C-x C-e with the values

```
cd ../src
```

```
g++ rectangle3.cpp -o rectangle3
echo "10 14 15 12 20 30" | ./rectangle3


What is the kitchen's length? 10
What is the kitchen's width? 14
What is the bedroom's length? 15
What is the bedroom's width? 12
What is the den's length? 20
What is the den's width? 30
The total area of the three rooms is 920
```

21 Constructors

- What is a constructor?

  A constructor is a `public` member function that is automatically called when a `class` is created: `Rectangle rect;`
  It has the name as the `class`, and it does not `return` anything.

- Simple example:

```
Rectangle() {}; // inline definition (in .h)

Rectangle::Rectangle() {}; // external definition (in .cpp)
```

- Demo example:

```
#include <iostream>
using namespace std;

class Demo
{
  public:
  Demo(); // Constructor declaration
};

Demo::Demo()
{
```

```
  cout << "Welcome to the Demo constructor!\n";
}

int main()
{
  Demo hello;
  return 0;
}
```

```
Welcome to the Demo constructor!
```

22 PRACTICE Improving `Rectangle` with a new explicit constructor

- To change the code on Google Cloud Shell, use the `nano` editor.
- We're going to make two improvements:
  1. Make the code safer against invalid data entry (implementation).
  2. Provide a constructor that initializes a rectangle of zero width and length using an initializer list.
- We'll keep the specification and the implementation apart in a .h and a .cpp file, and we'll test with a Makefile on the shell.
- `Rectangle.h` (with `#include` guard) - add new default constructor:

```
// Specification file for the Rectangle class
#ifndef RECTANGLE_H
#define RECTANGLE_H
class Rectangle
{
private:
  double width;
  double length;
public:
  Rectangle();                  // constructor
  void setWidth(double);        // mutator: Width
  void setLength(double);       // mutator: Length
  double getWidth() const       // accessor: Width
```

```
  {return width;}
  double getLength() const      // accessor: Length
  {return length;}
  double getArea() const        // accessor: Area
  {return width * length;}
};
#endif
```

- `Rectangle.cpp` (with `#include "Rectangle.h"`) - with data check:

```
#include "Rectangle.h"
#include <iostream>
#include <cstdlib>
using namespace std;

//************************************************************
// The constructor initializes width and length to 0.0
//************************************************************
Rectangle::Rectangle()
//: width(0.0), length(0.0) {}  // initialized with a list
{
  width = 0.0;
  length = 0.0;
}
//************************************************************
// setWidth sets the value of the member variable width
//************************************************************
void Rectangle::setWidth(double w)
{
  if (w >= 0)
    width = w;
  else
    {
      cout << "Invalid width.\n";
      exit(EXIT_FAILURE);           // or: return 1;
    }
}
//************************************************************
// setLength sets the value of the member variable length
//************************************************************
```

```
void Rectangle::setLength(double l)
{
  if (l >= 0)
    length = l;
  else
    {
      cout << "Invalid length.\n";
      exit(EXIT_FAILURE);
    }
}
```

- `RectangleTest.cpp` (with `#include "Rectangle.h"`) - make a rectangle:

```
#include <iostream>
#include "Rectangle.h"
using namespace std;

int main()
{
  Rectangle box;
  // Display rectangle data
  cout << "Here is the rectangle's data:\n";
  cout << "Width: "  << box.getWidth()  << endl
       << "Length: " << box.getLength() << endl
       << "Area: "   << box.getArea()    << endl;

  return 0;
}
```

- Test the new setup on the command-line:

```
cd ../src
g++ -c Rectangle.cpp -o Rectangle.o # produce Rectangle.o
g++ RectangleTest.cpp Rectangle.o -o rect # produce executable
./rect


Here is the rectangle's data:
Width: 0
Length: 0
Area: 0
```

- From C++11, you can also initialize a member variable in-place, namely when the variables are defined:

```
class Rectangle
{
  private:
    double width = 0.0;
    double length = 0.0;
};
```

23 Passing Arguments to Constructors

- A constructor can have parameters and can accept arguments when an object is created.
- Example: Here is a constructor with parameters for the `Rectangle` class.

  1. `Rectangle.h` - A `Rectangle` with undefined `width` and `length`

     ```
     Rectangle(double,double);
     ```

  2. `Rectangle.cpp` - A `Rectangle` with defined `width` and `length`

     ```
     Rectangle::Rectangle(double w, double l)
     {
        width = w;
        length = l;
     }
     ```

     Or with an initializer list:

     ```
     Rectangle::Rectangle(double w, double l) :
       width(w), length(l);
     ```

  3. `RectangleTest.cpp`: Creating a `Rectangle` with `width` and `length`

     ```
     Rectangle box(10.0, 12.0); // creates 10 x 12 rectangle
     ```

- You can also pass **default arguments** to the constructor. In the example, a default tax `rate` of 5% is written into the constructor.

  In `Sale.h` for a `Sale` class:

  ```
  Sale(double cost, double rate=0.05); // constructor
  ```

In `main`:

```
Sale itemSale(cost); // The Sale item has rate = 0.05
```

- What if a constructor has default arguments for all its parameters?

  Then it can be called without explicit arguments - it becomes the default constructor.

In `Rectangle.h`:xs

```
Rectangle(double w=0, double l=0); // Rectangle declaration
```

In `RectangleTest.cpp`:

```
// This works if either there is no default constructor, or
// your default constructor's parameters are all initialized.
Rectangle box;
```

- What if all constructors of a `class` require arguments?

  Then there is no default constructor. Now if you don't create with passing arguments, you get an error.

24 The Default Constructor and the RAII object lifecycle

- What is a destructor?

  A destructor is a member function that is automatically called when an object is destroyed (i.e. its memory is returned).

- When is an object destroyed?

  When it goes out of scope. Examples: at the end of a function if the objects are all local, or when they are deleted with `delete` after having been created with `new`.

- A demo example:

```
#include <iostream>
using namespace std;

// class interface
class Demo {
  public:
  Demo();  // default constructor
  ~Demo(); // default destructor
};
// class implementation
Demo::Demo() {cout << "Welcome to the constructor!\n";}
Demo::~Demo() {cout << "The destructor is now running!\n";}

// main function
// demonstrates an object with a constructor and a destructor
int main()
{
  Demo demoObject;
  return 0;
}
```

```
Welcome to the constructor!
The destructor is now running!
```

- What is the RAII principle?

  RAII = "Resource Allocation Is Initialization":

  1. The constructor initializes the resources
  2. The destructor releases them when the object is ready to die.

  = Object have a lifecycle.

  As a programmer you have control over life and death!